# Text Based Prediction of Book Review Popularity

**Goodreads Poetry mini project | UE24CS352A Machine Learning**

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Semester 5, Section I | PES University

### Problem statement

We study why some Goodreads book reviews receive a relatively large share of attention. For each review, engagement is its recorded votes plus comments. We divide that number by the total engagement of reviews for the same book and label the review popular if the share is strictly greater than 0.02. The task is to classify that label from reviewer, book and writing features. It measures relative engagement in an observed snapshot, not whether a review expresses positive sentiment or how many likes a new review will receive.

### Dataset

We use the Goodreads Poetry reviews and Poetry book metadata published with the UCSD Book Graph dataset. The review file supplies text, rating, date, reviewer and votes/comments; the book file supplies average rating and the metadata review count. Starting from 154,555 reviews, we exclude invalid negative or zero engagement, require at least 10 reviews in the book metadata and at least 60 total votes/comments for a book, and retain reviews identified as English by fastText with confidence of at least 0.90. This leaves 9,663 reviews from 462 books and 6,336 users. Of these, 2,604 (26.95%) have the popular label. Book engagement totals are computed before filtering by language or book eligibility.

### Approach

We compare four progressively richer feature sets: A contains four metadata features (user activity, review age, user rating and the difference from the book average); B adds eight writing and sentiment features; C adds bag-of-words counts to B; and D adds TF-IDF weights to B. C and D use at most 2,000 terms. Direct engagement counts and the derived share are excluded from every model input.

Books are held out across the training, validation and test partitions to prevent reviews of the same book appearing in multiple partitions. They contain 7,374, 1,184 and 1,105 reviews respectively. Text vocabulary and numeric scaling are fitted on training rows. For each feature set, we compare logistic regression, XGBoost and a small multilayer perceptron. Full and undersampled training are compared for A and B; C and D use undersampled training, for 18 experiments in total. Hyperparameters and the final model are chosen by validation ROC-AUC; the held-out test books are used for reporting.

## Implementation and evaluation

The Colab notebook handles data validation and filtering, feature extraction, book-disjoint splitting, model fitting, evaluation and result export. We report ROC-AUC, accuracy, precision, recall and specificity, alongside a majority-class baseline. The notebook also examines logistic-regression coefficients, XGBoost split counts, sampled neural-network SHAP values, score/engagement correlations and prediction-error groups. A supplementary text-only TF-IDF plus logistic-regression model powers the fresh-review demo; it is separate from the selected metadata-and-text experiments.

## Results from the recorded run

| Model and evaluation | Test ROC-AUC | Accuracy | Recall | Specificity |
|---|---|---|---|---|
| Majority-class baseline | 0.500 | 0.686 | 0.000 | 1.000 |
| XGBoost B, full training (selected) | 0.768 | 0.753 | 0.398 | 0.916 |
| XGBoost B, balanced training | 0.766 | 0.678 | 0.735 | 0.652 |
| Text-only demo model | 0.695 | 0.627 | 0.703 | 0.592 |

The full-training XGBoost B model had the highest validation ROC-AUC (0.729), so it was selected before interpreting test results. On 1,105 test reviews it produced 138 true positives, 64 false positives, 209 false negatives and 694 true negatives. Balancing training improves recall but lowers accuracy and specificity. The text-only demo has different inputs and should not be presented as the selected XGBoost model.

## Conclusion and limitations

Metadata and writing features distinguish above-threshold engagement better than the majority baseline on held-out Poetry books, but the selected model misses many popular reviews at a 0.5 threshold. User activity and review length are associated with model predictions; feature importance is descriptive, not causal. The data cover one genre and exclude zero-engagement reviews. User activity and review age are measured from a later snapshot, while labels use book-level totals, so this is retrospective classification rather than a publication-time forecast. The original multi-genre paper used different data, splits and models; these results are not its numerical reproduction.

## References and reproducibility

Daly, B. Text-Based Prediction of Book Review Popularity. Stanford STATS229, 2021. Reference code: github.com/bridgetdaly/goodreads_ML (commit 5297102c). Dataset: mengtingwan.github.io/data/goodreads.html. Our code and recorded outputs: github.com/Manaswi-Guntupalli/ML_Mini_Project_Book_Review_Popularity.